# SIMATS ENGINEERING
## ASSESSMENT TOOL 3: Mock Interview

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| **COURSE NAME:** | Cloud Computing and Big Data Analytics | **COURSE CODE:** | CSA1522 |

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| Register Number | 192525251 |
| Name | S.Blesssika |

**COURSE OUTCOME COVERED**
**CO5:** *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 25**
Question Count: 10

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**Code of Conduct**
I _S.Blessika (192525251)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ___S.Blesika______

**Assessment Tasks**
**Mock Interview**
Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

**Mock Interview Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Technical Knowledge | 25% | Shows deep and accurate technical understanding | Good technical understanding | Basic understanding | Weak understanding |
| Problem Solving | 20% | Responds with strong architecture-level reasoning | Reasonable reasoning | Basic reasoning | Weak reasoning |
| Communication | 20% | Answers are confident, precise, and professional | Mostly clear communication | Moderate clarity | Weak communication |
| Practical Awareness | 20% | Connects concepts to real-world implementation well | Some practical relevance | Limited practical relevance | No practical linkage |
| Confidence and Professionalism | 15% | Highly confident and professional | Generally professional | Moderately professional | Low confidence and professionalism |